All of the orbitals in a given shell have the same value for their _____ quantum number.

Select one:

- Secondary (Angular) (L)
- Spin (ms)
- Principal (n)
- Magnetic (mL)

How many unpaired electrons are found in V^{3+} (atomic number = 23)

Select one:

- 0
- 1
- 3
- 4

in the reaction

$$AsF_3 + CCl_4 \Longrightarrow AsCl_3 + CCl_2F_2,$$

If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 74.3%, how many grams of CCl_2F_2 (121 g/mol?) were actually obtained

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- 151
- 203
- 303
- 167
- 273

In the reaction

$$AsF_3 + CCl_4 \Longrightarrow AsCl_3 + CCl_2F_2,$$

If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 74.3%, how many grams of CCl_2F_2 were actually obtained?

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303

203

167

273

151

How many different orientations in space do s-orbitals have?

Select one:

- 5
- 2
- 7
- 1
- 3

A compound was found to contain 33.0% Si and 67.0% F, by mass. If the molar mass of the

compound is 170.0 g/mol, find its molecular formula.

(Si = 28 , F =19 g/mol)

Select one:

- SiF_3
- Si_2F_6
- Si_3F_9
- SiF_2

Which of the following is ?not an empirical formula

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- C_4H_8O
- $C_2N_2H_8$
- Sb_2S_3
- $BeCr_2O_7$
- MoO_2Cl_2

empirical formula SF_6?

S =32, F =19 g/mol

Select one:

- 29.67%
- 36.00%
- 21.95%
- 78.08%

The correct coefficient that balanced The O_2 in the chemical equation for the reaction below.

$$C_3H_8 + O_2 \Longrightarrow CO_2 + H_2O$$

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12

19

14

7

5

:chemical equation

$$C_4H_4 + 5\,O_2 \;\Longrightarrow\; 4\,CO_2 + .2\,H_2O$$

If 0.36 moles of C_4H_4 are allowed to react with 2.82 moles of O_2, what is the mass of water ?produced (in grams)

molar mass of water =18 g/mol

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- 13.20
- 19.64
- 12.96
- 65.50
- 11.02

$AsF_3 + CCl_4 \Longrightarrow AsCl_3 + CCl_2F_2$,

If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 74.3%, grams of CCl_2F_2 (121 g/mol) were actually obtained?

Select one:

- 273
- 167
- 303
- 151
- 203

In the reaction

$AsF_3 + CCl_4 \Longrightarrow AsCl_3 + CCl_2F_{2'}$

If the theoretical yield of CCl_2F_2 was 1.68 moles, and the percent yield in the reaction was 82.2 %, how many grams of CCl_2F_2 (121 g/mol) were actually obtained?

Select one:

- 167
- 203
- 273
- 151
- 303

below.

$$C_3H_8 + O_2 \Longrightarrow CO_2 + H_2O$$

Select one:

- 19
- 12
- 14
- 5
- 7

A compound was found to contain 33.0% Si and 67.0% F, by mass. If the molar mass of the compound is 170.0 g/mol, find its molecular formula.

(Si = 28 , F =19 g/mol)

Select one:

- Si_2F_6
- SiF_2
- SiF_3
- Si_2F_4

The lowest energy "f" subshell is : ____f

Select one:

- 4
- 2
- 5
- 3
- 1

Arrange the following atoms according to decrease in their ionization energies

Al (At.no. = 13)

Ga (At.no. = 31)

Mg (At.no. = 12)

Ar (At.no. = 18)

Select one:

- Ar >Mg> Al > Ga
- Mg > Al> Ga > Ar
- Ga> Mg > Al > Ar
- Ar >Al> Mg > Ga
- Al> Mg > Ar> Ga

Which of the following is not an empirical formula?

Select one:

- MoO_2Cl_2
- C_4H_8O
- $C_2N_2H_8$
- $BeCr_2O_7$
- Sb_2S_3

How many molecules of (C_4H_8O) are present in

2.00×10^{-8} mol of (C_4H_8O)?

Avogadro's number = 6.022×10^{23}

C = 12, H=1, O=16 g/mol

Select one:

- 1.67×10^{13}
- 1.20×10^{16}
- 1.20×10^{15}
- 1.67×10^{16}

Down the group (from top to bottom) the electron affinity _____

Select one:

- remains constant because effective nuclear charge is constant
- increases due to increase of number of shells
- decreases then increases
- decreases because number of shells increase
- increase because effective nuclear charge increases

Avogadro's number = 6.022×10^{23}

C = 12, H=1, O=16 g/mol

Select one:

- 0.64 mol
- 0.58 mol
- 0.78 mol
- 0.064 mol

How many different orientations in space do s-orbitals have?

Select one:

- 5
- 2
- 7
- 1
- 3